Esha Kar

Senior Full Stack AI-Native Developer | Next.js | Angular | Nest.js | AI Integration

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Started with a BA in English, then moved deliberately into software engineering — building technical foundations independently while working full-time and completing a BCA in parallel. That path taught me how to learn fast and go deep rather than stay within what I already know. Now a Senior Full Stack AI-Native Developer owning products end-to-end — architecture, hosting, domains, CDN, and AI integration — across ed-tech, OTT, and fintech, including real-time SDKs (LiveKit, Amazon Chime) for 1M+ user products.

Work Experience

Senior Full Stack AI-Native Developer — *House of Jana Ventures* Sep 2026 - Present

Haircraft — E-Commerce Platform (built from scratch) [Live ↗]

- Owned the project **end-to-end** — architecture, hosting, domain, VPS, and Cloudflare CDN, as sole full-stack owner.
- Built the storefront in **Next.js** and admin panel in **Angular**, backed by a **Nest.js** API layer, with AI capabilities integrated across the platform.

Jana Human Hair — Corporate Website & Platform Migration [Live ↗]

- Migrated site, domain, and DNS **off vendor hosting** to **Next.js** on **Vercel** + **Cloudflare**.

Full Stack Developer — *AscentSpark Pvt. Ltd.* Nov 2025 - Sep 2026

Tutopia Web App — Ed-Tech Platform (1M+ downloads) [Live ↗]

- Designed end-to-end **subscription and KYC payout flow** using Next.js and Laravel APIs, enabling tutor wallet operations for 1M+ users.
- Established **Quick Live 1:1 tutoring sessions** with real-time wallet-based booking and joining flow.
- Adopted **Amazon Chime SDK** for large-scale live classrooms, achieving 99.9% session reliability with raise-hand and moderation features.
- Reduced redundant API calls by **35%** and improved application performance by **30%** via TanStack Query caching.

AI Spoken Learning — Real-time AI Agent for Spoken English [Live ↗]

- Integrated **LiveKit SDK** as real-time communication layer, cutting session join time by 40% vs prior Chime approach.
- Produced frontend for **live AI evaluation** — audio-level indicators, transcription display, and instant feedback scoring.
- Implemented full **session lifecycle**: room creation, participant join, AI agent connection, and graceful teardown.

SSEN Samsung TV — OTT Platform for Sports & Football [Live ↗]

- Developed a **React OTT application** for Samsung Tizen Smart TVs with D-pad navigation and live football streaming pipelines.

FreshFishBasket — E-Commerce Platform [Live ↗]

- Integrated **Razorpay payment gateway** end-to-end — order creation, payment capture, webhooks, and refund processing.

Software Development Intern — *ITJOBXS* Jan 2025 - Oct 2025

- Fixed **20+ API bugs**, reducing average server response time by 25%.
- Implemented **Redis rate limiting and session caching** from scratch, cutting abuse by 40%.
- Optimised 5+ MongoDB queries with indexing and aggregation, saving 120ms per query.

Frontend Developer — *People Maketh* May 2024 - Dec 2024

Internal Data Verification System [Live ↗]

- Consolidated **10+ REST APIs** into React dashboards, reducing manual verification time by 25%.
- Created a reusable component library, cutting UI rebuild time by 35% across the team.

Projects

Agentic AI Evaluation Platform [GitHub ↗] Live ↗

- **FastAPI** + **Next.js** app benchmarking **Claude**/**Gemini** agents on answers and tool use.
- Rule-based + **LLM-as-judge** scoring; **Docker** and **GitHub Actions CI**.

AI Resume Analyzer & Interview Prep Live ↗

- Launched a **Gemini API**-powered app generating role-specific interview questions, cutting prep time by 60%.
- Scoring engine returns ranked skill-gap results in under 3s; handles 300+ requests at sub-500ms.

MERN Notes App Live ↗

- Handled **500+ daily requests**; Redis caching cut load by 30%, indexing cut latency 20%.

Avatar Development Live ↗

- Built an interactive **avatar rendering** application in React, exploring dynamic customization and real-time preview.

Technical Skills

LANGUAGES

JavaScript (ES6+), TypeScript, Python

FRONTEND

React.js, Next.js, Angular, Tailwind CSS, Three.js, GSAP

BACKEND & APIS

Node.js, Nest.js, Express.js, Django, FastAPI, REST, JWT

INFRA & DEVOPS

VPS, Cloudflare (CDN/DNS), Domain Management, CI/CD Pipelines

REAL-TIME & SDKS

LiveKit SDK, Amazon Chime SDK, Firebase Realtime DB, WebSockets

PAYMENTS & INTEGRATIONS

Razorpay, Payment Gateway Integration, Webhooks, E-Commerce APIs

AI & DATA

RAG Pipelines, LangChain, Gemini API, ChatGPT API, Vector DBs

DATABASES

MongoDB, PostgreSQL, MySQL, Redis

PRACTICES

TanStack Query, Query Optimisation, Component Architecture, Production Debugging

TOOLS

Docker, Git, GitHub, Postman, Figma, Netlify

Education

BCA — Amity University

2024 – 2027 | CGPA: 8.76 | alongside full-time work

BA English — Univ. of Kalyani

2020 – 2023 | CGPA: 7.5

Achievements

- **5-star HackerRank** rating in algorithms and data structures — top percentile.

Freelance

IN PROGRESS

LabNest labnest-self.vercel.app ↗

Lab management platform built with **Next.js** and **React**. Currently under active development — handling frontend architecture, component design, and API integration.